Module 12-5 to 12-10 #1

Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **29**

Instructor Name: **Bodanske, Daniel**

Question 1 of 29

Fill in the blank with a number to make the expression a perfect square.

$$x^2 + 6x + \square$$

$$x^2 + 6x + \square$$

Question 2 of 29

Solve the quadratic equation by completing the square.

$$x^2 - 2x - 6 = 0$$

First, choose the appropriate form and fill in the blanks with the correct numbers.
Then, solve the equation. Round your answer to the nearest hundredth.
If there is more than one solution, separate them with commas.

Form:

○ $(x + ___)^2 = ___$

○ $(x - ___)^2 = ___$

Solution:

$x = ___$

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Question 3 of 29

Consider the following quadratic function.

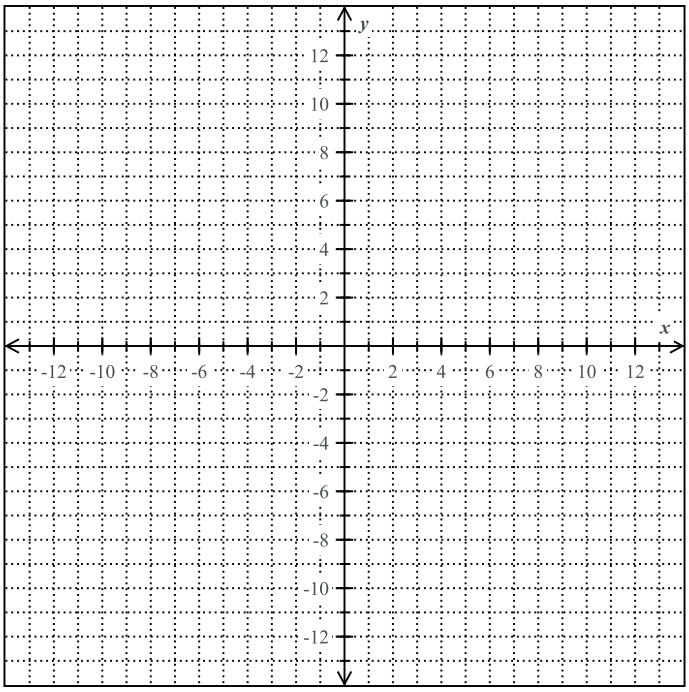
$$f(x) = 2x^2 + 20x + 49$$

(a) Write the equation in the form $f(x) = a(x-h)^2 + k$. Then give the vertex of its graph.

Writing in the form specified: $f(x) =$

Vertex:

(b) Graph the function.



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Question 4 of 29

Consider the following quadratic function.

$$f(x) = -x^2 - 4x + 3$$

(a) Write the equation in the form $f(x) = (x-h)^2 + k$ or the form $f(x) = -(x-h)^2 + k$.

$f(x) =$ _____

(b) Find the equation of the axis of symmetry.

Equation of axis of symmetry:

_____

(c) Does the function have a minimum or maximum value (output)?

- Minimum
- Maximum

(d) For what input, x, does the minimum or maximum value occur?

$x =$ _____

(e) What is the function's minimum or maximum value?

_____

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Question 5 of 29

A ball is thrown vertically upward. After t seconds, its height h (in feet) is given by the function $h(t) = 76t - 16t^2$. What is the maximum height that the ball will reach?

Do not round your answer.

Height: _____ feet

Question 6 of 29

Use the quadratic formula to solve for x.

$$6x^2 - 9x = -1$$

Round your answer to the nearest hundredth.

$x =$ ☐

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Question 7 of 29

Let's derive the formula to find the solutions of a quadratic equation, known as the *quadratic formula*.

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Given: Here is the standard form of a quadratic equation, where a, b, and c are coefficients. We want to solve this equation for x.

$$ax^2 + bx + c = 0$$

Step 1: Rewrite the equation so that all the x-terms are on the left side and other terms are on the right side.

$$ax^2 + bx = -\underline{\quad}$$

Step 2: Rewrite the equation so that the leading coefficient is 1.

$$x^2 + \frac{b}{\underline{\quad}}x = -\frac{c}{\underline{\quad}}$$

Step 3: Rewrite the equation so that the left side is a perfect square trinomial.

$$x^2 + \frac{b}{a}x + \left(\frac{\underline{\quad}}{\underline{\quad}}\right)^2 = \left(\frac{b}{\underline{\quad}}\right)^2 - \frac{c}{a}$$

Step 4: Factor the perfect square.

$$\left(x + \frac{\underline{\quad}}{\underline{\quad}}\right)^2 = \left(\frac{b}{\underline{\quad}}\right)^2 - \frac{c}{a}$$

Step 5: Simplify the right side of the equation so there is a common denominator.

$$\left(x + \frac{\underline{\quad}}{\underline{\quad}}\right)^2 = \frac{b^2 - \underline{\quad}}{4a^2}$$

Step 6: Take the square root of both sides of the equation.

$$x + \frac{\underline{\quad}}{\underline{\quad}} = \pm\sqrt{\frac{b^2 - \underline{\quad}}{4a^2}}$$

Step 7: Simplify the right side.

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{\underline{\quad}}$$

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Step 8: Isolate x on the left side and simplify.

$$x = \frac{- \underline{\quad} \pm \sqrt{b^2 - 4ac}}{\underline{\quad}}$$

Question 8 of 29

Discriminant:

Number of real solutions:

Compute the value of the discriminant and give the number of real solutions of the quadratic equation.

$$5x^2 - x + 4 = 0$$

Discriminant: _____

Number of real solutions: _____

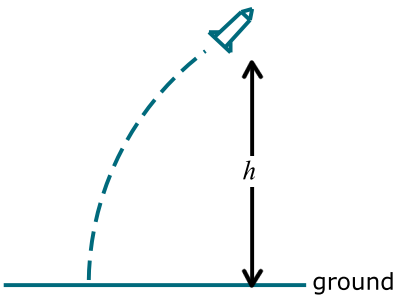
Question 9 of 29

A model rocket is launched with an initial upward velocity of 65 m/s. The rocket's height h (in meters) after t seconds is given by the following.

$$h = 65t - 5t^2$$

Find all values of t for which the rocket's height is 30 meters.

Round your answer(s) to the nearest hundredth.
(If there is more than one answer, use the "or" button.)



$t =$ _____ seconds

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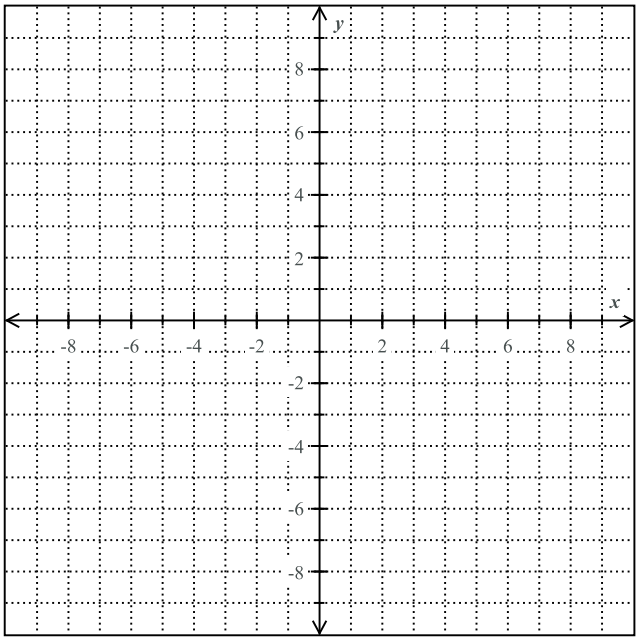
Question 10 of 29

Graph the system below.
Then write its solution.

$$\begin{cases} y = -2x^2 \\ y = -2x - 4 \end{cases}$$



Question 11 of 29

Solve the following system of equations.

$$\begin{cases} y = x^2 - 6 \\ 8x + y = -18 \end{cases}$$

Question 12 of 29

Use a graphing calculator to solve the system of equations.

$$y = -3x^2 - 18x - 25$$
$$y = 2x + 6$$

Round to the nearest hundredth.

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Question 13 of 29

For each function, state whether it is linear.

Function 1	Function 2
$\{(4, 2), (8, 2), (12, 2), (16, 2)\}$	$\{(3, 3), (5, 4), (7, 5), (9, 6)\}$
○ Linear ○ Not linear	○ Linear ○ Not linear

Function 3

x	y
-2	-5
1	-4
4	-1
7	3

○ Linear
○ Not linear

Function 4

x	y
-3	-1
-2	-6
-1	-11
0	-16

○ Linear
○ Not linear

Question 14 of 29

A table of values of a linear function is shown below.

x	-1	0	1	2	3
y	4	7	10	13	16

Find the slope and y-intercept of the function's graph, and find the equation for the function.

slope: _____

y-intercept: _____

equation: _____

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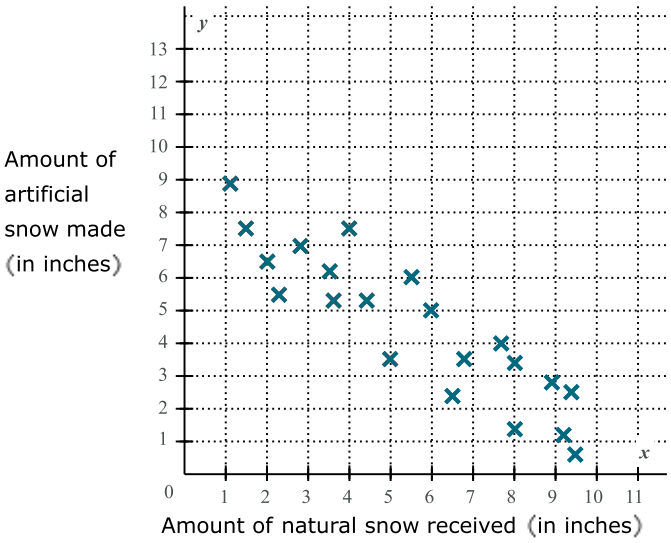
Question 15 of 29

The data points on the scatter plot below show the amount of natural snow received and the amount of artificial snow made by a ski resort for each of 21 weeks.

Draw the line of best fit for these data points. (It doesn't have to be the exact line of best fit. Just draw your best approximation.)



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Question 16 of 29

The data points show an ocean trench's depth y (in feet) at a distance x from a marked buoy (in feet).

Each figure has the same data points.
However, each figure has a different curve fitting the data.
The equation for each curve is also shown.

Answer the questions that follow.

Figure 1

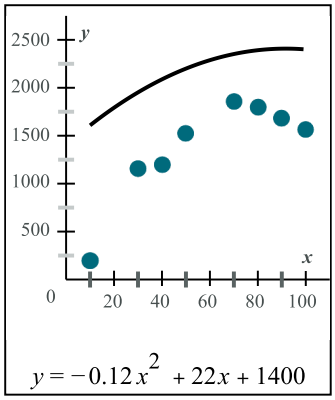


Figure 2

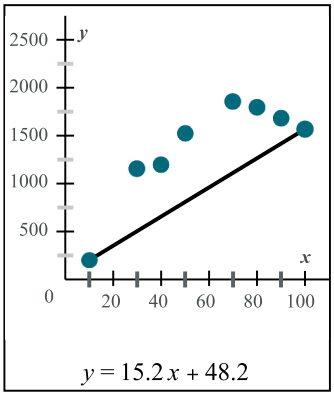
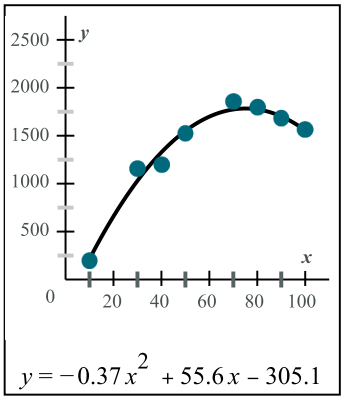


Figure 3



(a) Which curve fits the data best?

○ Figure 1 ○ Figure 2 ○ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the depth of the trench 60 feet from the buoy. Give an exact answer, not a rounded approximation.

_____ feet

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Question 17 of 29

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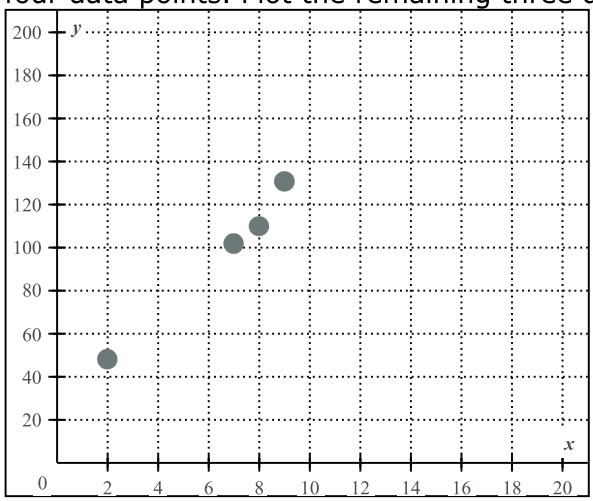
Ivan collected the following data from a random sample.

x	2	7	8	9	13	16	17
y	48	102	110	131	180	181	182

Complete the parts below to help Ivan find a relationship between x and y for his dataset.

(a) To see if there are any trends in the data, Ivan wants to create a scatter plot. He plotted the first four data points. Plot the remaining three data points.



(b) The data could have an exponential trend or quadratic trend. So, Ivan wants to determine which equation (exponential or quadratic) could better fit the data. The outputs for an exponential regression model and a quadratic regression model are given below. Use this to answer the question below.

x	2	7	8	9	13	16	17
y	48	102	110	131	180	181	182

ExpReg: $y = a\,b^x$	**QuadReg:** $y = a\,x^2 + b\,x + c$
$a = 51.529$	$a = -0.329$
$b = 1.088$	$b = 15.841$
	$c = 13.321$
$R^2 = 0.881$	$R^2 = 0.972$

Which regression model more closely fits the data?

○ ExpReg

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○ QuadReg

(c) Write the equation of the model that best fits the data. Write the values in your answer to 3 decimal places.

$y =$ _____

(d) Use the equation from part (c) to help Ivan predict the value of y when $x = 12$. Write your answer to the nearest whole number.

_____

Question 18 of 29

The function h is defined by the following rule.

$$h(x) = 7^x$$

Find $h(x)$ for each x-value in the table.

x	$h(x)$
-1	_____
0	_____
1	_____
2	_____
3	_____

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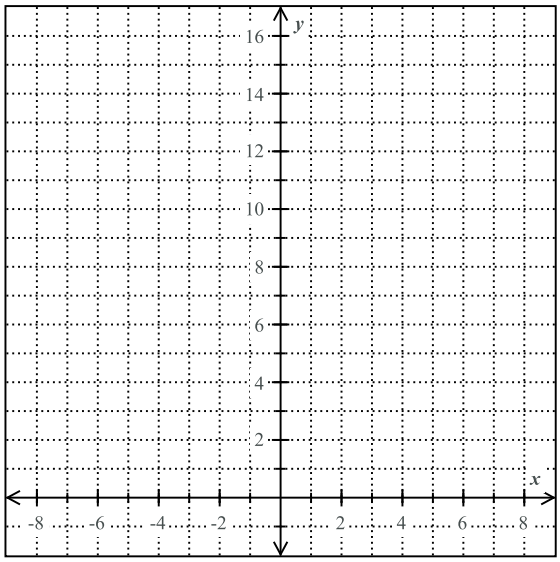
Question 19 of 29

Graph the exponential function.

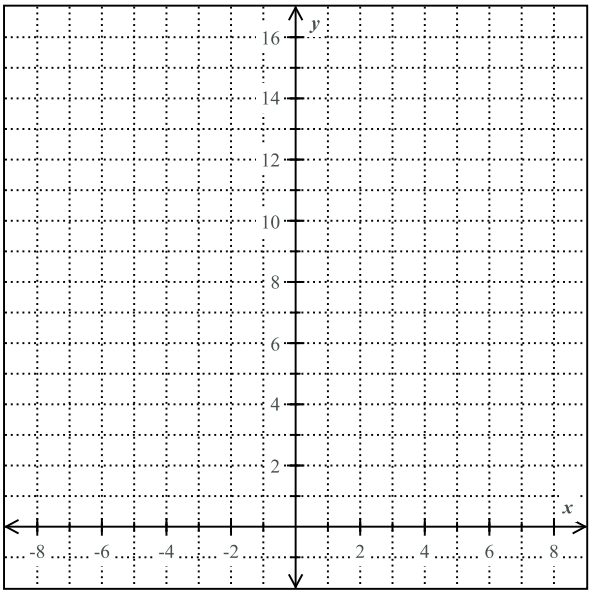
$$f(x) = 2^x$$



Question 20 of 29

Graph the exponential function.

$$f(x) = \left(\frac{8}{3}\right)^x$$



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Question 21 of 29

The table of ordered pairs (x, y) gives an exponential function.
Write an equation for the function.

x	y
-1	$\frac{1}{27}$
0	$\frac{1}{3}$
1	3
2	27

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Question 22 of 29

The data points show the area y (in square kilometers) that a certain forest covers after being inhabited for a time x (in years).

Each figure has the same data points.
However, each figure has a different curve fitting the data.
The equation for each curve is also shown.

Answer the questions that follow.

Figure 1

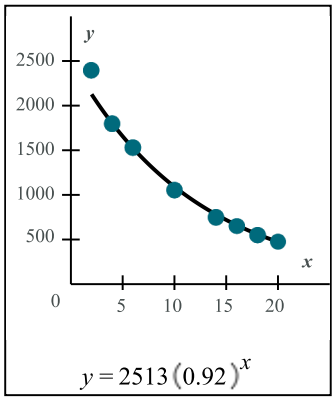


Figure 2

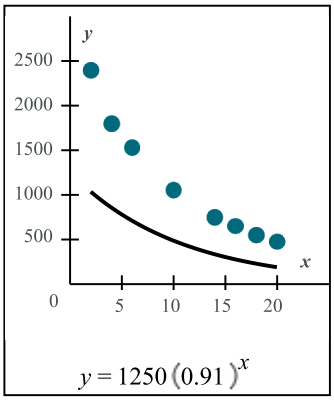
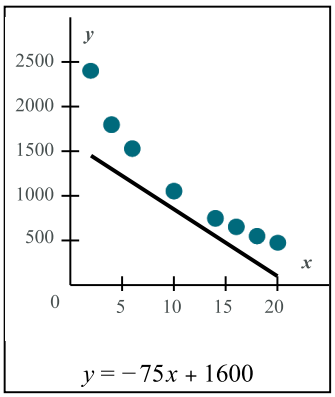


Figure 3



(a) Which curve fits the data best?

○ Figure 1 ○ Figure 2 ○ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the area that the forest covers after it is inhabited for 12 years. Round your answer to the nearest hundredth.

_____ square kilometers

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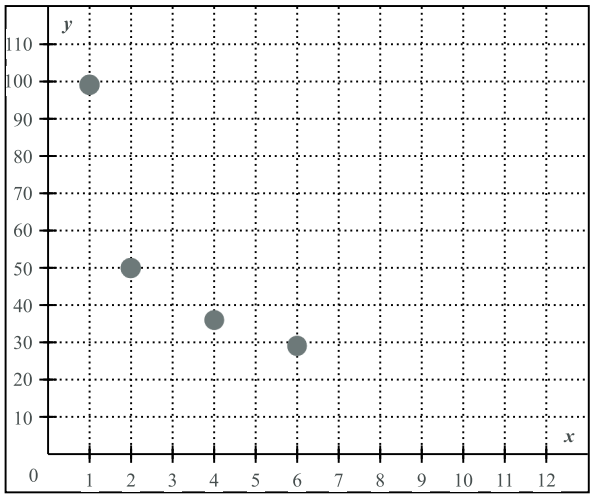
Question 23 of 29

Manuel collected the following data from a random sample.

x	1	2	4	6	7	8	10
y	99	50	36	29	16	8	6

Complete the parts below to help Manuel find a relationship between x and y for his dataset.

(a) To see if there are any trends in the data, Manuel wants to create a scatter plot. He plotted the first four data points. Plot the remaining three data points.



(b) The data could have a linear trend or an exponential trend. So, Manuel wants to determine which equation (linear or exponential) could better fit the data. The outputs for a linear regression model and an exponential regression model are given below. Use this to answer the question below.

(c) Write the equation of the model that best fits the data. Write the values in your answer to 3 decimal places.

(d) Use the equation from part (c) to help Manuel predict the value of y when $x = 5$. Write your answer to the nearest whole number.

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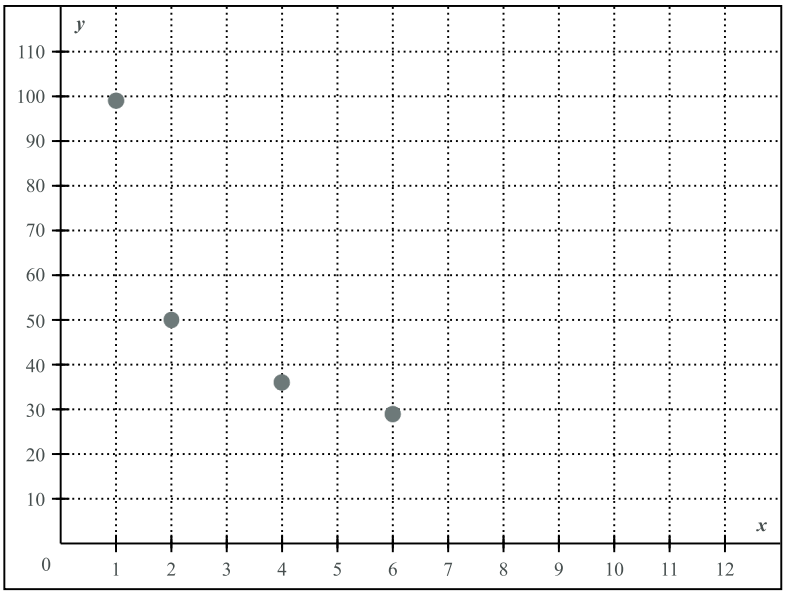
(a) To see if there are any trends in the data, Manuel wants to create a scatter plot. He plotted the first four data points. Plot the remaining three data points.



(b) The data could have a linear trend or an exponential trend. So, Manuel wants to determine which equation (linear or exponential) could better fit the data. The outputs for a linear regression model and an exponential regression model are given below. Use this to answer the question below.

x	1	2	4	6	7	8	10
y	99	50	36	29	16	8	6

LinReg: $y = ax + b$	**ExpReg:** $y = a\,b^x$
$a =$	$a =$
$b =$	$b =$
$R^2 =$	$R^2 =$

Which regression model more closely fits the data?

- LinReg (linear)
- ExpReg (exponential)

(c) Write the equation of the model that best fits the data. Write the values in your answer to 3 decimal places.

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$y =$ _____

(d) Use the equation from part (c) to help Manuel predict the value of y when $x = 5$. Write your answer to the nearest whole number.

_____

Question 24 of 29

The first three terms of a geometric sequence are as follows.

$$81, 27, 9$$

Find the next two terms of this sequence.

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Question 25 of 29

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For each function, state whether it is linear, quadratic, or exponential.

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(a)

x	y
2	-15
3	-16
4	-20
5	-27
6	-37

Classify the function.

- Linear
- Quadratic
- Exponential
- None of the above

(b)

x	y
4	112
5	56
6	28
7	14
8	7

Classify the function.

- Linear
- Quadratic
- Exponential
- None of the above

(c)

x	y
1	7

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2	13
3	19
4	25
5	31

Classify the function.

- Linear
- Quadratic
- Exponential
- None of the above

Question 26 of 29

A small publishing company is releasing a new book. The production costs will include a one-time fixed cost for editing and an additional cost for each book printed. The total production cost C (in dollars) is given by the function $C = 19.95N + 650$, where N is the number of books.

The total revenue earned (in dollars) from selling the books is given by the function $R = 33.70N$.

Let P be the profit made (in dollars). Write an equation relating P to N. Simplify your answer as much as possible.

Question 27 of 29

Suppose that the functions r and s are defined for all real numbers x as follows.

$$r(x) = 4x^3$$
$$s(x) = x^2$$

Find the following.

$(r - s)(x) =$ _____

$(r + s)(x) =$ _____

$(r \cdot s)(2) =$ _____

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Question 28 of 29

The functions r and s are defined as follows.

$$r(x) = 2x - 1$$
$$s(x) = -2x^2 - 1$$

Find the value of $s(r(5))$.

$s(r(5)) =$ $\square$

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Question 29 of 29

For each pair of functions f and g below, find $f(g(x))$ and $g(f(x))$.
Then, determine whether f and g are inverses of each other.

Simplify your answers as much as possible.
(Assume that your expressions are defined for all x in the domain of the composition.
You do *not* have to indicate the domain.)

(a) $f(x)=x-4$ $g(x)=x+4$ $f(g(x))=$ $g(f(x))=$ - f and g are inverses of each other - f and g are *not* inverses of each other	(b) $f(x)=\frac{1}{6x}$ $g(x)=-\frac{1}{6x}$ $f(g(x))=$ $g(f(x))=$ - f and g are inverses of each other - f and g are *not* inverses of each other

(a) $f(x)=x-4$

$g(x)=x+4$

Find the compositions.

$f(g(x)) =$ _____

$g(f(x)) =$ _____

Are f and g inverses of each other?

- ○ f and g are inverses of each other
- ○ f and g are *not* inverses of each other

(b) $f(x)=\frac{1}{6x}$, $x\neq 0$

$g(x)=-\frac{1}{6x}$, $x\neq 0$

Find the compositions.

$f(g(x)) =$ _____

$g(f(x)) =$ _____

Are f and g inverses of each other?

- ○ f and g are inverses of each other
- ○ f and g are *not* inverses of each other

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **29**

Question 1 of 29

$x^2 + 6x + 9$

Question 2 of 29

Form:

- ○ $(x + \square)^2 = \square$
- ◉ $(x - 1)^2 = 7$

Solution:

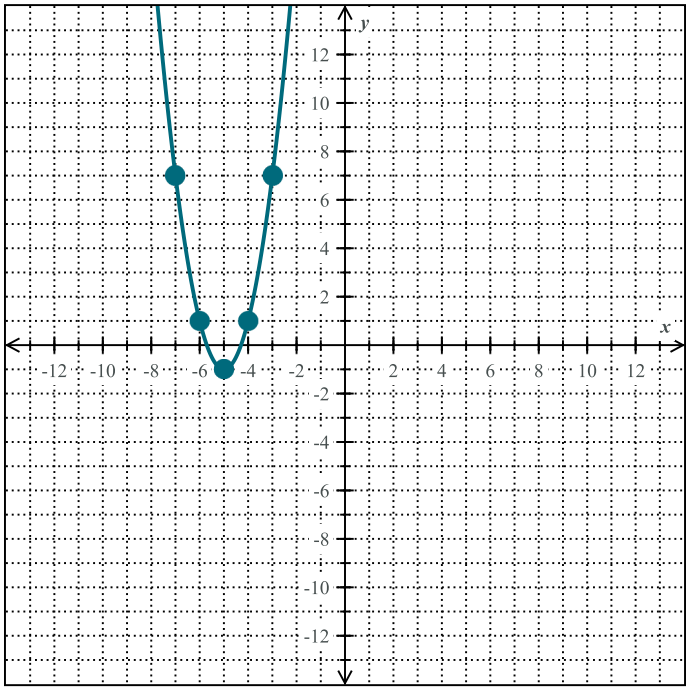
$x = 3.65, -1.65$

Question 3 of 29

(a) Writing in the form specified: $f(x) = 2(x + 5)^2 - 1$

Vertex: $(-5, -1)$

(b)



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Question 4 of 29

(a) $f(x) = -(x + 2)^2 + 7$

(b) Equation of axis of symmetry:
$x = -2$

(c) ○ Minimum ◉ Maximum

(d) $x = -2$

(e) 7

Question 5 of 29

Height: 90.25 feet

Question 6 of 29

$x = 1.38, 0.12$

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Question 7 of 29

Step 1: $$ax^2 + bx = -c$$

Step 2: $$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Step 3: $$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a}$$

Step 4: $$\left(x + \frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a}$$

Step 5: $$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Step 6: $$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

Step 7: $$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

Step 8: $$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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Question 8 of 29

Discriminant: -79

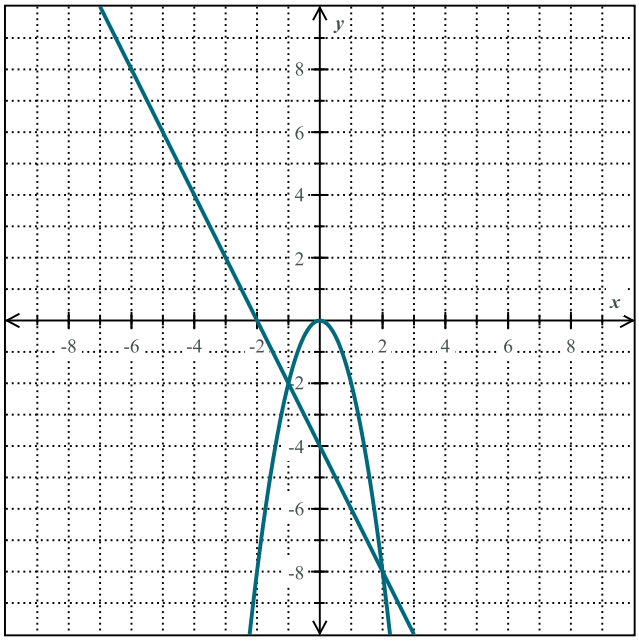
Number of real solutions: 0

Question 9 of 29

$t = 0.48$ seconds

or $t = 12.52$ seconds

Question 10 of 29



Solution(s):

$(x, y) = (2, -8)$

or $(x, y) = (-1, -2)$

Question 11 of 29

$(x, y) = (-2, -2)$

or $(x, y) = (-6, 30)$

Question 12 of 29

$(x, y) = (-4.22, -2.43)$

or $(x, y) = (-2.45, 1.10)$

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Question 13 of 29

Function 1	Function 2
$\{(4, 2), (8, 2), (12, 2), (16, 2)\}$ ◉ Linear ○ Not linear	$\{(3, 3), (5, 4), (7, 5), (9, 6)\}$ ◉ Linear ○ Not linear
Function 3	**Function 4**
x | y -2 | -5 1 | -4 4 | -1 7 | 3 ○ Linear ◉ Not linear	x | y -3 | -1 -2 | -6 -1 | -11 0 | -16 ◉ Linear ○ Not linear

Question 14 of 29

slope: 3

y-intercept: 7

equation: $y = 3x + 7$

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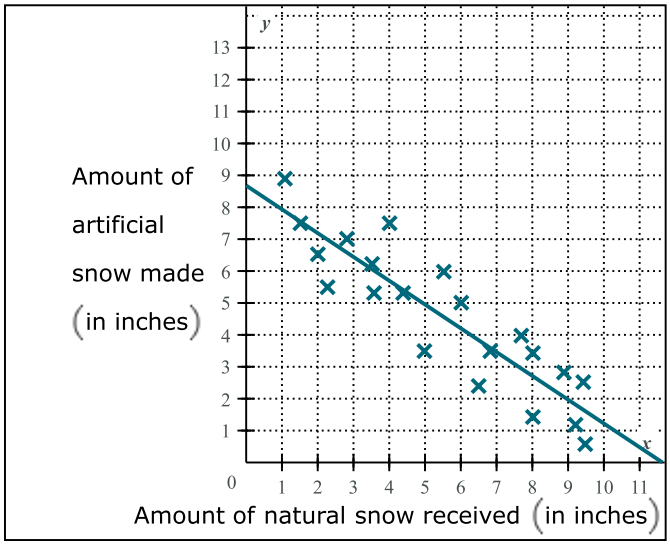
Question 15 of 29



Question 16 of 29

(a) Which curve fits the data best?

○ Figure 1 ○ Figure 2 ◉ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the depth of the trench 60 feet from the buoy. Give an exact answer, not a rounded approximation.

1698.9 feet

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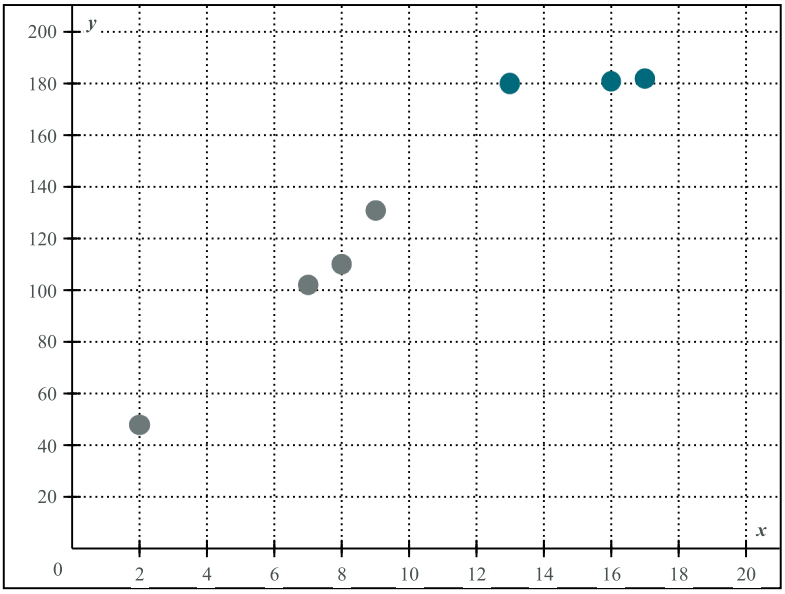
(a)



(b) ExpReg (exponential)

QuadReg (quadratic)

(c) $y = -0.329x^2 + 15.841x + 13.321$

(d) $y = 156$

Question 18 of 29

x	$h(x)$
-1	$\frac{1}{7}$
0	1
1	7
2	49
3	343

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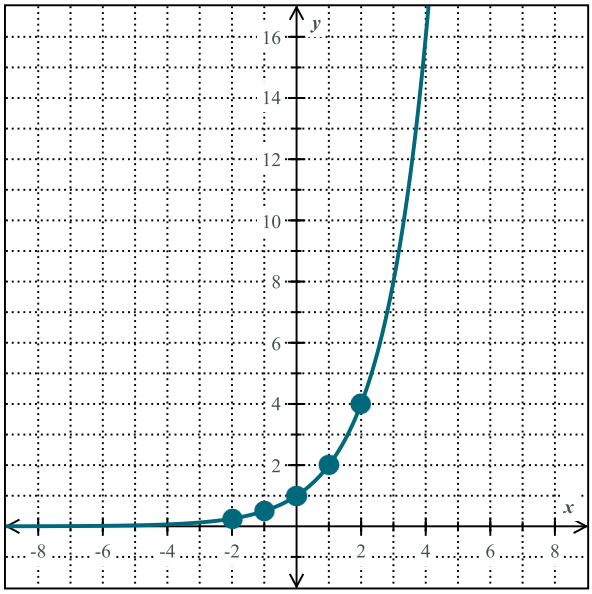
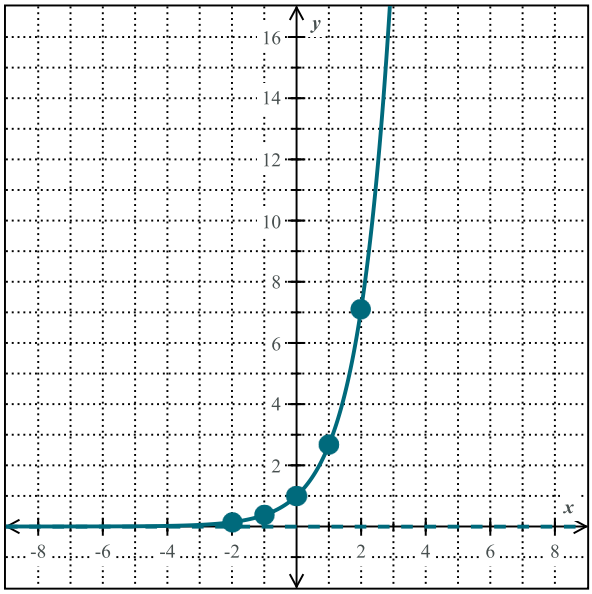
Question 19 of 29



Question 20 of 29



Question 21 of 29

$$y = \frac{1}{3} \cdot 9^x$$

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Question 22 of 29

(a) Which curve fits the data best?

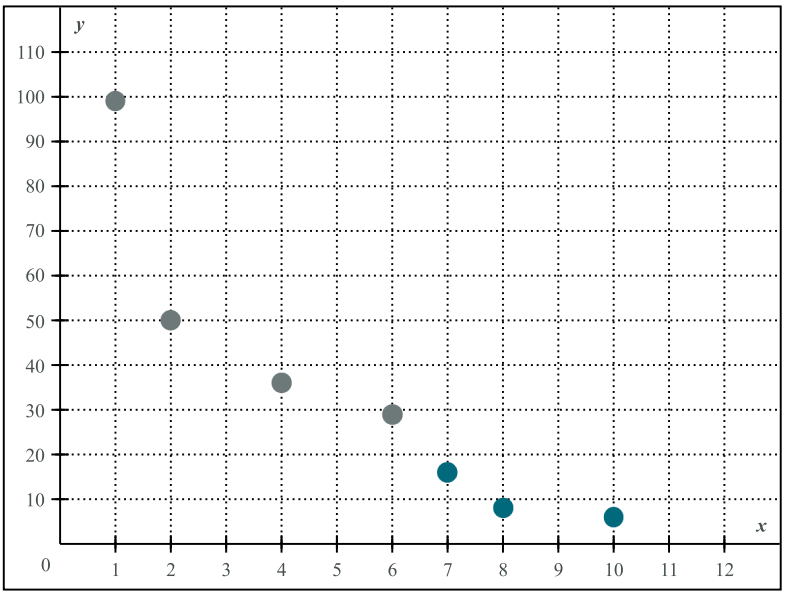
- (●) Figure 1
- ( ) Figure 2
- ( ) Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the area that the forest covers after it is inhabited for 12 years. Round your answer to the nearest hundredth.

923.95 square kilometers

Question 23 of 29

(a)



(b)
- ( ) LinReg (linear)
- (●) ExpReg (exponential)

(c) $y = 119.291\left(0.741\right)^{x}$

(d) $y = 27$

Question 24 of 29

$81, 27, 9, 3, 1$

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Question 25 of 29

(a)
- ○ Linear
- ◉ Quadratic
- ○ Exponential
- ○ None of the above

(b)
- ○ Linear
- ○ Quadratic
- ◉ Exponential
- ○ None of the above

(c)
- ◉ Linear
- ○ Quadratic
- ○ Exponential
- ○ None of the above

Question 26 of 29

$P = 13.75N - 650$

Question 27 of 29

$$(r - s)(x) = 4x^3 - x^2$$

$$(r + s)(x) = 4x^3 + x^2$$

$$(r \cdot s)(2) = 128$$

Question 28 of 29

$s(r(5)) = -163$

Note: The value $s(r(5))$ is called the composition of the function s with r at $x = 5$.

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Question 29 of 29

(a) $f(x) = x - 4$

$g(x) = x + 4$

$f(g(x)) = x$

$g(f(x)) = x$

- (●) f and g are inverses of each other
- ( ) f and g are *not* inverses of each other

(b) $f(x) = \frac{1}{6x}, x \neq 0$

$g(x) = -\frac{1}{6x}, x \neq 0$

$f(g(x)) = -x$

$g(f(x)) = -x$

- ( ) f and g are inverses of each other
- (●) f and g are *not* inverses of each other

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